

3-9-2023

PHYSICS

Date:

Q 1. (4) Presbyopia

Q 2. (2) Repulsion

Q 3. (4) Virtual, erect, and same size as you

Q 4. (2) Below The image of fish

Q 5. $P = +1.5 \text{ D}$

$$P = \frac{1}{f}$$

$$f = \frac{1}{P} \Rightarrow f = \frac{1}{1.5} \Rightarrow \frac{10}{15}$$

$$= +0.606 \text{ m}$$

The lens prescribed is convex lens, which is converging lens

Q 6. $I = 5 \text{ A}$, $t = 30 \text{ mins}$

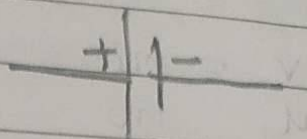
$$= 30 \times 60$$

$$= 1800 \text{ s}$$

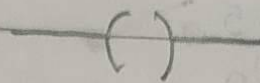
$$I = \frac{Q}{t}$$

$$5 = \frac{Q}{1800} \Rightarrow \underline{Q = 9000 \text{ C}}$$

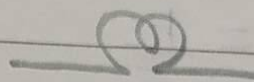
Q7. Electric cell



Switch open



Electric bulb



Battery



A wire joint



Q8. $h_o = 5 \text{ cm}$

$f = 12 \text{ cm}$

$u = -20 \text{ cm}$

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

$$\frac{1}{12} = \frac{1}{v} - \left(\frac{1}{-20} \right)$$

$$\frac{1}{12} = \frac{1}{v} + \frac{1}{20} \Rightarrow \frac{1}{v} = \frac{1}{12} - \frac{1}{20}$$

$$\Rightarrow \frac{1}{v} = \frac{1 \times 5}{12 \times 5} - \frac{1 \times (-3)}{20 \times (-3)}$$

$$\Rightarrow \frac{1}{v} = \frac{5 + 3}{60}$$

$$\Rightarrow \frac{1}{v} = \frac{8}{60} = 1.5 \text{ cm}$$

$$m = \frac{v}{u} = \frac{h_i}{h_o}$$

$$m = \frac{7.5}{-20}$$

$$\frac{7.5}{-20} = \frac{h_i}{5}$$

$$\frac{37.5}{-20} = h_i$$

$$h_i = -1.875 \text{ cm}$$

∴ The image is real inverted and magnified.

Q.9. The potential difference between the two ends of the conductor (V) = $10\text{V} - 5\text{V} = 5\text{V}$
 $W = \frac{5\text{J}}{1\text{C}}$

$$V = \frac{W}{Q}$$

$$Q = \frac{W}{V} = \frac{5\text{J}}{5\text{V}} = \underline{1\text{C}}$$

Q10. a) Pole: Pole is the centre of reflecting surface of the spherical mirror.

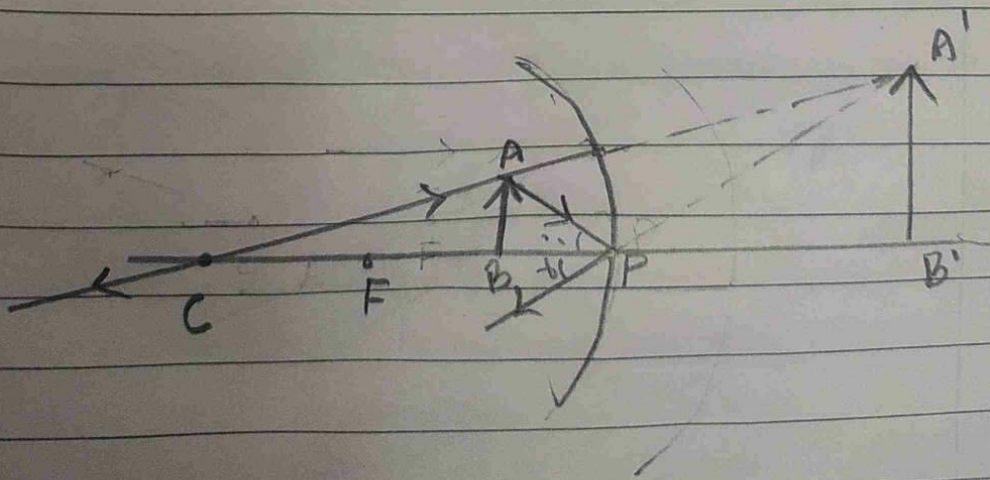
b) Radius of curvature is the radius of the hollow sphere.

c) Principle axis is a straight line passing through centre of curvature and pole.

d) Focus is a point on the principle axis at which the light rays meet after reflection.

The range of distance of the candle flame from the mirror should be less than 20 cm i.e. it should be placed between the focus and pole of the mirror.

The other characters of the image are virtual, erect and magnified.



Q 11. a) Since blue colour has a shorter wavelength and red colour has a longer wavelength, the red colour is able to reach our eye after the atmospheric scattering of light. Hence, the sun appears reddish early in the morning.

b) The ability of eye lens to focus near and far objects clearly on retina by adjusting its focal length is called power of accommodation.

c) The least distance of distinct vision of a normal human eye is 25 cm.